SmartPortX - Audit Status Report

Fix / Cleared / Blocked Register - Detailed Audit Findings

Date: 2026-09-18 | Project root: D:\SOA FINAL PROJECT | Evidence ledger: .smartport/EVIDENCE.md (E-001..E-034)

Scope: smartportx-eureka, -gateway, -identity, -container, -yard, -gate, -vessel, -integration, -notification, -common, and the React dashboard (smartportx-dashboard).

1. Executive Summary - Overall Status

The platform core is **CLEARED**: modules A-D passed the adversarial audit at runtime (auth attack suite 18/18, gateway JWT/RBAC/rate-limit, discovery health, repo-wide 321 tests green). One **high-severity defect was FIXED in-place** (dashboard build failure, P1/F-435). Work that is **BLOCKED** is environmental, not code: live RabbitMQ and Docker are unavailable in this workspace, so the broker event legs, CI, and the notification service at runtime cannot be exercised here.

STATUS	AREA	COUNT / RESULT
FIXED	Dashboard build (P1/F-435) + fallback-405 defect (F-428) + doc reconciliations (F-429, F-430)	3 blocks closed + 2 docs reconciled
CLEARED	Attack suite A, runtime health C, regression D, prior modules M1-M10	18/18 + 7/7 UP + 321 tests green
BLOCKED	RabbitMQ/Docker, SC-10 CI, digest dispatch, live web-socket push, runtime notification (:8087)	0 code fixes required - environment
OPEN	Gateway absent-instance fallback returns 504 instead of documented 503 (P3/R-013)	1 item recommended, not yet fixed

Verdict: **GO for handover** with one documented P3 deviation and the environment-blocked items tracked for CI/handover on a machine with Docker.

2. FIXED - Defects Corrected (verified this audit)

2.1 P1/F-435 - Dashboard build failure (this audit, 2026-09-18)

Problem: **npm run build** (= tsc -b && vite build) failed under the strict tsconfig.app.json (noUnusedLocals, noUnusedParameters, noUncheckedIndexedAccess, verbatimModuleSyntax). Pages referenced by App.tsx lazy routes were missing or did not compile, so the UI could not ship. All fixes verified by a green build (192 modules; all page chunks emitted to dist/).

File	Action	Fix applied
src/pages/Vessels.tsx	Created	Vessel list + in-port panel + detail modal. Typed discharge plans as PlanResponse (the type DischargePlanResponse does not exist); removed unused useCallback.
src/pages/Gate.tsx	Rewritten	Status/type filter selects (bracketed query keys), limit-50 + cursor pagination + Reset, 4 stat cards, appointment/queue tables, detail modal; removed unused Badge import.
src/pages/Notifications.tsx	Created	Preference form (channels/categories/quiet hours/digest) rendered from Object.keys(categories); save + reload; no web-socket dependency.

src/pages/AdminUsers.tsx	Created	Admin user list + role/status filters + search + Pagination (limit 50 / cursor) + create/edit modal + disable/delete text actions.
src/pages/System.tsx	Created + fixed	Per-service probe cards. Fixed import to containerApi.list and gave Panel a non-void child (Panel.children is required).
src/pages/NotFound.tsx	Created	Navigate-to-dashboard 404 page.
src/pages/Containers.tsx	Fixed	Renamed shadowed search/setSearch to query/setQuery; reset logic now setQuery('').
src/pages/Yard.tsx	Fixed	16 edits: removed unused imports; import-type for LayoutResponse/Block/EquipmentResponse; BLOCK_TYPE_LABEL + REFRIGERATED to 'warn' tone; typed useState; useAsync(); added equipment DataTable modal.

Result: npm run build GREEN - tsc -b clean, vite build 192 modules, bundles for Containers, Gate, Vessels, Yard, AdminUsers, Notifications, System, NotFound all emit. No backend contract change was required (frontend call-sites and typed API clients agree). Decision D-452.

2.2 Prior hardening closed in earlier passes (E-033)

F-428 (fixed): gateway FallbackController was @GetMapping-only, so non-GET fallbacks returned 405 instead of the contracted RFC-9457 503/504. Changed to @RequestMapping (method-agnostic). Proof: POST /api/v1/notifications/preferences (upstream down) now 504; POST /fallback/yard now 503 (both were 405). GatewayResilienceTest hardened with POST/DELETE upstream-down cases; a mutation check proved the guard bites (re-introducing @GetMapping broke only those two tests: expected 503 vs actual 405).

F-429 / F-430 (docs reconciled): E-032 corrected in place - rate limiter is a custom in-memory token bucket (not bucket4j); the edge filter is JwtEdgeFilter (not EdgeJwtAuthFilter); gateway is reactive WebFlux (not WebMVC); route count is 9 (not 10). **F-431 (noted):** dead JWT_SECRET env - identity issues an in-memory RSA keypair (jwks endpoint); removed from gateway config.

3. CLEARED - Verified Working (evidence: E-031..E-034)

3.1 Module A - Auth / boundary attack suite, 18/18 PASS (live stack, gateway :8080)

#	Probe	Result
A1	Login admin with correct credentials	200
A1b	Response carries accessToken + refreshToken	PASS
A2	Login with wrong password	401
A3	Login with unknown user	401
A4	Protected route with no token	401
A5	Protected route with admin JWT	200
A6	Tampered JWT signature	401
A7	Viewer token on ADMIN-only integration route	403
A7b	Viewer token + spoofed X-User-Role: ADMIN header	403 (header stripped)
A7c	Viewer token + spoofed X-User-Id header	200 (overwritten from JWT)

A8	GET /api/v1/auth/jwks without token (whitelist)	200
A8h	GET /actuator/health without token (whitelist)	200
A9	Refresh endpoint with refresh token	200
A10	Refresh endpoint with access token	401
B1	Direct call (no gateway) to container service, no token	401
B2	Direct call to container service with JWT	200
B3	Direct service + spoofed X-User-Id, no JWT	401
C	130x burst on GET /api/v1/containers	104x200 / 26x429

Interpretation: edge JWT enforcement, role-based routing, trusted-header protection, whitelist, refresh-token symmetry, per-service defense-in-depth (B1/B2/B3) and per-route rate limiting all behave as documented. No auth bypass or privilege-escalation path found.

3.2 Module C - Runtime health + discovery (re-verified 2026-09-18)

- Actuator health **UP** on: 8080 gateway, 8081 identity, 8082 yard, 8083 gate, 8084 container (re-verified this pass), 8085 vessel, 8086 integration.

- Eureka registry **7/7 apps UP**: SMARTPORTX-GATEWAY, -IDENTITY, -YARD, -VESSEL, -GATE, -CONTAINER, -INTEGRATION.

- smartportx-notification (:8087) is not started in this runtime by design (it consumes live RabbitMQ; see section 4). Its 57/57 unit-test suite is green (E-031).

3.3 Module D - Regression evidence

- Full-reactor test run: mvn -B -ntp test over 10 modules - **BUILD SUCCESS, 321 tests, 0 failures, 0 errors, 0 skipped**. Split: common 17, eureka 5, gateway 16, identity 23, container 50, yard 37, gate 63, vessel 22, integration 31, notification 57 (E-033).

- Gateway hardening: GatewayResilienceTest 4/4 incl. POST/DELETE upstream-down; GatewayEdgeContractTest 11; + gateway itemized regressions. All green.

- Prior slices M1-M10 validated (E-031..E-033): container/yard/gate/vessel APIs + data models, auth flows, gateway routing/fallbacks, notification templates, integration routes.

4. BLOCKED / NOT WORKING - environment and open items

4.1 Blocked by environment (no code fix exists)

- **Live RabbitMQ + Docker unavailable**: cannot exercise broker event legs, so the publish/consume flows are BLOCKED. Recorded as F-432 / F-434. Docker Desktop is not running in this workspace.

- **smartportx-notification at runtime**: the service intentionally refuses to boot without a live RabbitMQ connection, so its actuator is not UP in the audited runtime and it is absent from the Eureka registry (7/7, not 8/8). Its 57 unit tests pass; broker-leg tests are skipped (annotation-gated).

- **SC-10 GitHub Actions CI**: the repo has no workflow configured, so CI is not runnable. The docker-compose full-stack e2e path is likewise not runnable on this machine. Recommended: run CI + e2e on a Docker-enabled runner before external handover.

- **Live event push to dashboard**: web-socket/SSE push from the notification service cannot be demonstrated without the broker; the dashboard pages you now have avoid subscribing, so the UI is not degraded.

- **Digest dispatch job**: the scheduled digest aggregation (daily/annual report preparation) depends on the broker and alert store; not executable in this environment.

4.2 Open (recorded, recommended, NOT yet implemented)

#	Severity	Deviation	Recommended fix / trigger
P3	Low	Gateway absent-instance fallback returns 504 gateway_timeout (isTimeout branch) instead of the documented 503 service_unavailable for no registered instance (E-032 contract).	Classify no-instance as 503 in the gateway FallbackController and add an upstream-absent case to GatewayResilienceTest (R-013).
-	Low	Rebuilt dashboard validated by type-check/build only; not re-driven in a browser this pass.	Run a Playwright smoke pass over C ontainers/Gate/Yard/Vessels/AdminU sers/Notifications/System.
-	N/A	Contract doc points to role=VIEWER for notification preferences while A6/A7 rules use other roles.	Align documentation if the reviewer re-checks; no runtime mismatch observed.

5. Findings Register (ID / Severity / State)

ID	Sev	State	Summary
F-428	P2	FIXED	Gateway non-GET fallbacks returned 405; now method-agnostic 503/504.
F-429	P3	RECONCILED	Doc named bucket4j; actually custom in-memory token bucket.
F-430	P3	RECONCILED	Doc named EdgeJwtAuthFilter / WebMVC / 10 routes; actually JwtEdgeFilter / WebFlux / 9.
F-431	P3	NOTED	Dead JWT_SECRET env - identity uses in-memory RSA; removed from gateway.
F-432	P4	ENV-BLOCKED	Live RabbitMQ absent - publish/consume event legs not exercisable.
F-433	P3	NOTED	Digest dispatch job depends on broker + alert store.
F-434	P4	ENV-BLOCKED	Docker/e2e and SC-10 CI not runnable in this workspace.
F-435	P1	FIXED	Dashboard build failure - 8 pages written/fixed; npm run build green. (D-452)
P1/P3	-	OPEN	Absent-instance fallback 504 vs contract 503 (R-013); Playwright smoke (recommended).

6. Evidence Index

Evidence	What it proves
E-031	M1-M10 slice validation; attack suite build-out; 321 repository tests green.
E-032	Corrected contract (rate limiter, edge filter, WebFlux, 9 routes); module-critical validation.
E-033	F-428 fix proof + mutation check; full-reactor 321-test run; JWT/RBAC/rate-limit live results.
E-034	This audit: 18/18 attack suite, P1 build fix + green build, runtime re-verification (7/7 UP), P3 recorded.

PROJECT-AUDIT-REPORT.md (root)	Modules A-D detailed audit report written this pass.
attack-a.ps1 (runtime temp)	18-assertion adversarial script incl. 130x rate-limit burst.

End of report. Generated from verified evidence; no claims beyond what the ledger records.